CONSOLIDATED REPLICATION PROMPT

Genetic Algorithm for the Travelling Salesman Problem

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Course	CSE3121 — Artificial Intelligence and Soft Computing
Submitted to	Dr. Neha Chaudhary
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Document purpose

This consolidated, refined prompt specifies the intended implementation and deliverables for the GA–TSP algorithmic implementation component. It brings the notebook, report, and presentation requirements into one place so the work can be reproduced and evaluated consistently.

> The original, unedited session prompts are documented separately in PROMPT_HISTORY.pdf. This document is a post-project specification, not a verbatim conversation transcript.

1. Role and Overall Goal

Complete the “Algorithmic Implementation of AI approaches with Viva” component (10 marks) for CSE3121 by implementing a Genetic Algorithm (GA) for the Travelling Salesman Problem (TSP) over 15 Indian cities. Validate the GA against the exact optimum and produce three deliverables: a Jupyter notebook, a LaTeX report, and a Beamer presentation.

The work should be reproducible, correct, and explainable in a viva. The prompt explicitly says not to include a viva-questions section in the notebook.

2. Deliverable 1 — Jupyter Notebook

`GA_TSP_Algorithmic_Implementation.ipynb`

Problem data. Use Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad, Pune, Jaipur, Lucknow, Bhopal, Patna, Chandigarh, Guwahati, and Kochi. Represent each by latitude/longitude and build a distance matrix using the haversine formula with Earth radius R = 6371 km.

GA design table. Explain the permutation chromosome, fitness = 1 / tour length, tournament selection (k = 5), Order Crossover (OX1, probability 0.9), segment-inversion mutation (probability 0.2), elitism (best 2 unchanged), population 150, 300 generations, and fixed random seeds.

Implementation from scratch. Do not use GA libraries. Implement tour_length, tournament_select, order_crossover, inversion_mutation, and genetic_algorithm. Check that operators preserve valid permutations.

Run and visualise. Print the best tour and runtime. Plot the route on a longitude/latitude map with the start city marked. Plot convergence using best-so-far and population-average curves.

Exact validation. Implement Held–Karp dynamic programming, O(n² 2■). Report the optimum, the GA gap to it, and the optimal route. Expected optimum: 7478.9 km. Note that one GA run can stall in a local optimum.

Experiments. Save figures to figures/ and tables as CSV. Required experiments are detailed in Section 3.

Worked operator examples. Show an eight-city Order Crossover example: P1 = 1..8, P2 = [3,7,5,1,6,8,2,4], slice positions 2 to 4. Include an inversion-mutation example.

Complexity and limitations. GA: O(G·P·n) time and O(P·n) space. Held–Karp: O(n² 2■) time and O(n 2■) space. Discuss no optimality guarantee, seed/parameter dependence, and premature convergence.

Conclusion. Draw conclusions only from the generated experimental results. Print computed results instead of hard-coding result text.

3. Experimental Plan

Experiment	Required setup	Output
GA vs. baselines	Random search with equal evaluation budget (150 × 300), averaged over 3 seeds; nearest neighbour; GA over 10 seeds.	Mean and best tour length; number of GA runs reaching optimum.
Mutation rate	Rates {0, 0.05, 0.2, 0.5, 0.9}; 8 seeds each; 200 generations.	Box plot with optimum reference line.
Population size	Sizes {10, 30, 100, 300}; equal budget 45,000 evaluations; 8 seeds each.	Box plot comparing final tour lengths.

All plots and CSV tables should be generated by the notebook and saved in figures/. Fixed seeds are specified to make runs repeatable.

4. Deliverable 2 — LaTeX Report

`report.tex` compiled to `report.pdf`

Use an A4 article layout at 11 pt. The title page must include the college logo (college_logo.jpg), department, course name and code, project title, student name and roll number, “Submitted to: Dr. Neha Chaudhary”, and academic year 2026–27.

Required report structure

- Abstract and table of contents.
- Introduction: problem statement with objective formula, motivation for using a GA, and project objectives.
- Methodology: city data, GA design table, encoding, fitness, selection, crossover, mutation, elitism, and pseudocode.
- Implementation: key code excerpts.
- Results: all notebook figures and tables, each with a short interpretation.
- Complexity and Limitations.
- Conclusion.

Use figures saved by the notebook. Compile with `tectonic report.tex` or `pdflatex report.tex` twice, and fix compilation errors.

5. Deliverable 3 — Beamer Presentation

`beamer/presentation.tex` compiled to `presentation.pdf`

Use the Madrid theme with the beaver colour theme and aim for 15–17 slides. Load figures from ../figures/. The title slide must show the full student name, roll number, department, course, and supervisor; the full name must also appear in the footer.

Required slide topics

- Outline and TSP problem.
- Why a GA and project objectives.
- GA design table.
- Worked Order Crossover example.
- Exact optimum versus GA route and convergence.
- Baseline comparison table.
- Mutation-rate and population-size experiments.
- Complexity, limitations, and conclusion.

Use `\AtBeginSection` to repeat the outline. Every numerical result on slides must come from the notebook. Compile from the beamer directory using `tectonic presentation.tex`.

6. Quality and Reproducibility Requirements

- Use fixed random seeds; running all notebook cells should regenerate every figure and CSV.
- Use computed values in result text rather than hard-coded numbers.
- Report outcomes honestly: the prompt expects the GA to be near-optimal on average and find the optimum in about half of runs, while acknowledging a single-run local optimum.
- Verify by running and compiling deliverables. Record warnings or failures rather than hiding them.
- Keep the same style as existing files and avoid unnecessary extra files.

7. Expected Headline Results — Replication Check

These are expected values supplied in the original consolidated prompt. They are checkpoints for comparing a replication, not proof that a fresh run has independently reproduced them.

Method	Length (km)	Gap to optimum
Exact optimum (Held–Karp)	7478.9	0%
GA, best of 10 runs	7478.9	0%
GA, mean of 10 runs	7526.9	0.6%
Nearest neighbour	9169.4	22.6%
Random search, same budget	9930.4	32.8%

The prompt also specifies that the GA reaches the optimum in 5 of 10 runs. If a fresh execution differs, inspect the implementation, random seeds, environment, and evaluation protocol, then report observed results rather than forcing expected values.

8. Suggested Project Folder Structure

```
GA_TSP_Algorithmic_Implementation.ipynb
figures/
college_logo.jpg
report.tex
report.pdf
beamer/
    presentation.tex
    presentation.pdf
```

The notebook creates the figures directory. The Beamer presentation loads figures from one directory level above its own folder.

9. Final Replication Checklist

- ■ Run every notebook cell from a clean/restarted kernel.
- ■ Confirm all required figures and CSV files are regenerated.
- ■ Check that GA operators preserve valid permutations.
- ■ Compare computed results against Held–Karp and the baseline methods.
- ■ Compile the report and presentation from their specified directories.
- ■ Check that all figures load and slide numbers are notebook-derived.
- ■ Confirm student, roll number, course, and supervisor details in report and slides.
- ■ Document warnings, errors, or differences from expected results.

Source note: This PDF reformats the supplied REPLICATION_PROMPT.md into a structured specification for easier review and execution. Requirements and expected values are retained from that source; no independent project run is claimed here.